Practice Quiz: Perception (Mathematical Frameworks)

Name: __ ***Date:*** __

Part A: Multiple Choice

1. Variational free energy F can be decomposed as: A) F = Reward - Cost B) F = Complexity - Accuracy = D_KL[q||prior] - E_q[log P(o|s)] C) F = Input + Output D) F = Prior × Likelihood

2. For Gaussian models, minimizing free energy reduces to minimizing: A) The sum of all observations B) Precision-weighted prediction error: $\frac{1}{2} \pi(o - g(s))^2$ C) The number of neurons D) The distance between neurons

3. Precision π in the prediction error equation represents: A) How fast the brain processes information B) The reliability/confidence of a sensory channel — inverse variance C) The size of the prediction error D) The location of the observation

4. When precision is high, a prediction error: A) Is ignored B) Has a large influence on belief updating C) Becomes negative D) Doesn't exist

5. In hierarchical free energy minimization, prediction errors flow: A) Downward (from higher to lower levels) B) Upward (from lower to higher levels) C) Only sideways D) In no particular direction

6. The accuracy-complexity trade-off means: A) Maximize accuracy at all costs B) Good models explain data well (accuracy) while remaining as simple as possible (complexity) C) Complexity is always bad D) Accuracy is impossible to achieve

7. Gradient descent on free energy is the mathematical description of: A) Physical movement B) Belief updating — iteratively adjusting beliefs to reduce prediction error C) Muscle contraction D) Memory storage

Part B: Short Answer

1. You observe a temperature of 25°C. Your model predicts 20°C. The precision of the thermometer is $\pi = 2$. Compute the prediction error and the weighted prediction error ($\frac{1}{2} \pi(o - g(s))^2$). If precision drops to $\pi = 0.5$ (unreliable thermometer), recompute. What changes?

2. Explain how the equation F = Complexity - Accuracy unifies two desires: fitting the data and keeping the model simple. Give a perception example where the brain must balance these.

3. In a hierarchical model with 3 levels, explain what happens when a completely unexpected stimulus appears at the lowest level (e.g., a flying pig). How do prediction errors propagate up the hierarchy, and how might higher levels respond?